Notes: Brown, Martinsson, and Petersen (JF, 2013)

Law, Stock Markets, and Innovation

Contents

1 Big picture

- **Question.** Do legal rules and financial market development affect firm-level innovative investment, and if so, is the key channel access to stock market financing rather than access to credit?
- **Setting.** The paper studies about 5,300 publicly traded non-U.S. firms across 32 countries over 1990–2007 using Compustat Global and North America, merged with country-level measures of law, equity market access, and credit market development.
- **Core challenge.** A large literature links legal investor protections to stock market development and links finance to growth, but it is harder to show the specific firm-level real activity through which these relationships operate. The paper also needs to distinguish stock market access from general financial development and from debt market access.
- **Main conceptual contribution.** The paper argues that the most important firm-level channel is *R&D investment*. R&D is intangible, offers little collateral, and has very skewed payoffs, so debt contracts are poorly suited to finance it. By contrast, fixed capital is tangible and easier to pledge, so firms can more readily fund it with debt.
- **Main hypotheses.**
  - Better access to stock market financing should raise firm-level R&D intensity.
  - The effect should be strongest in small and young firms, because they are the most likely to need external finance.
  - Credit market development should matter much less for R&D than stock market access.
  - Stock market access should matter less for fixed investment than for R&D.
  - Legal rules that protect minority shareholders should affect R&D mainly through the availability of external equity financing.
- **Main empirical design.** The authors adapt the Rajan–Zingales (1998) difference-in-difference approach to the firm level. They interact an industry's technological dependence on external finance with country-level measures of equity-market or credit-market access, while controlling for country and industry fixed effects plus firm controls.
- **Main baseline result.** Stock market access has a positive and economically meaningful effect on R&D investment. Using country equity issues as the preferred measure, the baseline interaction coefficient is 0.413 and the implied high-minus-low differential in R&D intensity is 0.015.
- **Main heterogeneity result.** The R&D effect is concentrated in small and young firms. For small firms, the differential in R&D intensity is about 0.022 when comparing a highly externally dependent industry to a low-dependence industry in a high-stock-access country versus a low-stock-access country. This is about one-third of average small-firm R&D intensity.
- **Main contrast result.** Credit market development does *not* raise R&D investment, while stock market access has essentially no effect on fixed investment. Credit access has at most a modest effect on fixed capital spending.
- **Main mechanism result.** Strong investor protections are associated with more firm-level stock issuance, especially by small firms, and with R&D that is less tied to internal cash flow. This supports the idea that law matters because it changes access to external equity funding.

- **Main growth result.** Better stock market access is associated with more intangible assets and faster value-added and productivity growth for small firms, consistent with the idea that stock markets foster growth through innovation rather than through physical capital accumulation.
- **Economic takeaway.** Law and stock markets matter most for investments that are risky, intangible, and hard to collateralize. The paper therefore identifies a concrete channel through which legal institutions and equity-market development can affect innovation and long-run growth.

2 Data and measurement (key objects)

2.1 Sample construction

- The starting point is standardized financial-statement data for firms covered in Compustat Global and Compustat North America over 1990–2007.
- The paper excludes financial firms (SIC 6000–6999) and utilities (SIC 4900–4999).
- It keeps firms with fully consolidated statements, at least three nonmissing R&D observations, and nonmissing employment information.
- U.S. firms are excluded from the estimation sample because U.S. data are used to construct the benchmark measure of industry external-finance dependence.
- All firm-year ratios are Winsorized at the 1% level before constructing firm averages.
- The core analysis uses *firm averages over 1990–2007* rather than annual panel variation. The object of interest is the long-run link between financing environment and investment intensity.
- The final sample contains about 5,300 firms across 32 countries.

Interpretation. This is not a short-run cash-flow-sensitivity paper. The authors deliberately average over time so that stock market access is linked to long-run investment levels rather than to year-to-year financing shocks.

2.2 Why the paper focuses on firm-level evidence

- There is large heterogeneity across firms in dependence on external finance.
- Country-level R&D aggregates include universities and other organizations that are not central to the financing hypothesis.
- The paper wants to observe how law and finance affect real activity at the firm level, not just in macro aggregates.

Interpretation. The firm-level design is essential because the identifying idea comes from comparing firms in different industries with different external-finance needs inside the same country.

2.3 Firm-level outcome variables

The main firm variables are long-run firm averages:

- **R&D** = average of R&D expenditures divided by total assets.
- **Fixed investment** = average fixed investment divided by total assets.
- **Intangible assets** = average stock of intangible assets divided by net fixed assets.
- **Value-added growth** = average annual log change in operating income plus labor expense.
- **Productivity growth** = average revenue growth minus 0.3× growth in fixed assets minus 0.7× growth in employment.
- **Cash flow** = average after-tax income before extraordinary items plus depreciation and amortization plus R&D, all divided by total assets.
- **Stock issues** = average net equity issues divided by total assets.
- **Sales** = average net sales divided by total assets.
- **Sales growth** = average annual log change in sales.
- **Employment** = average number of employees.

Interpretation. The paper treats R&D, fixed investment, intangible accumulation, and growth as related but distinct margins. This is important because the paper's theory predicts very different effects across these outcomes.

2.4 Industry dependence on external finance

The Rajan–Zingales-style industry measure is:

$$ExternalDepend_j = \frac{\text{fixed investment} - \text{cash from operations}}{\text{fixed investment} + \text{R\&D}} \quad \text{for the median U.S. firm in industry } j, \tag{1}$$

where the numerator and denominator are summed over 1990–2007 before taking the ratio.

- The measure is computed for each two-digit SIC industry using U.S. firms.
- Scaling by fixed investment plus R&D is an important modification relative to standard Rajan–Zingales because R&D is expensed and therefore reduces cash flow directly.

Interpretation. The idea is that the United States provides the closest available benchmark for an industry's *technological* demand for external funds. Industries that naturally require outside finance should benefit more from better financial-market access.

2.5 Country-level finance variables

The paper distinguishes carefully between stock market access and credit market access.

- **Country equity issues**: the country average of the firm-level ratio of net stock issues to total assets.
- **IPOs/GDP**: value of IPOs relative to GDP, averaged over 1996–2000.
- **Accounting standards**: a measure of the comprehensiveness of corporate annual reports in 1990.

- **Country-weighted leverage**: value-weighted average debt-to-assets ratio across sampled firms in a country.
- **Credit/GDP**: private credit by deposit money banks divided by GDP, averaged over 1990–2007.

Interpretation.

- The first three variables are proxies for the access firms have to *external equity finance*.
- The last two are proxies for access to *debt finance*.
- A major theme of the paper is that broad development measures like stock market capitalization or private credit-to-GDP can be poor proxies for the actual funding available to listed firms, so the paper puts weight on measures tied more directly to observed financing behavior.

2.6 Country-level legal variables

The legal and institutional variables are:

- **Legal origin**: indicator for common-law origin.
- **Anti-self-dealing index (ASD)**: protection of minority shareholders against self-dealing and insider expropriation.
- **Enforcement**: extent to which private contracts are enforced.

Interpretation. These variables matter because the law-and-finance literature argues that they are powerful determinants of stock market development, especially the ability of outside equity investors to supply funds.

2.7 Firm size and age classifications

- A firm is **small** if its average employment is in the bottom 70% of sampled firms.
- A firm is **large** otherwise.
- A firm is **young** if it first appears in Compustat after 1990.
- A firm is **mature** if it is already in Compustat by 1990.

Interpretation. These splits proxy for ex ante dependence on external finance. Small and young firms are the natural place to look for stronger effects of equity-market access.

2.8 Descriptive facts from Table II

- In the full sample, mean R&D/assets is 0.053 and mean fixed investment/assets is 0.048, so R&D is quantitatively as important as physical capital spending.
- Small firms are much more R&D intensive than large firms: 0.066 versus 0.024.
- Small firms have lower cash flow and issue much more external equity than large firms.
- Mean stock issues/assets is 0.088 for small firms but only 0.012 for large firms.

- Mean employment is about 640 for small firms and 18,145 for large firms.

Interpretation. The summary statistics line up very closely with the paper's financing story. Small firms both invest more heavily in R&D and rely more on equity issuance, while large firms look much more internally financed.

2.9 Country coverage and appendix facts

- The sample covers roughly 75% of global non-U.S. R&D based on 2007 figures, and more than 90% once Russia and China are excluded.
- Japan contributes the largest share of firms, which is why later robustness tests drop the biggest contributor countries.
- The appendix table shows substantial cross-country variation in legal protections, equity issuance, accounting standards, leverage, and credit market depth.

Interpretation. The paper combines broad country coverage with enough within-country cross-industry structure to run a Rajan–Zingales design at the firm level.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism

The central mechanism is straightforward:

- R&D is intangible and has little collateral value.
- R&D also has highly skewed payoffs: many failures, but potentially very large upside when projects succeed.
- Standard debt contracts do not handle this payoff structure well because creditors capture downside protection but do not share equally in upside gains.
- External equity is therefore much better suited than debt for financing innovative investment.

Interpretation. Once this mechanism is accepted, the empirical predictions become sharp: stock market access should matter for R&D, but not much for fixed investment, and especially not in large mature firms that can fund more internally.

3.2 Baseline Rajan–Zingales regression

The paper's main specification is:

$$R\&D_{ijk} = \gamma_0 + \gamma_1(ExternalDepend_j \times FinDevelop_k) + \gamma_2 X_{ijk} + \lambda_k + \omega_j + \varepsilon_{ijk}, \tag{2}$$

where:

- $R\&D_{ijk}$ is firm i's long-run R&D/assets in industry j and country k;

- $ExternalDepend_j$ is industry j's technological dependence on external finance;
- $FinDevelop_k$ is a country-level measure of access to equity or debt finance;
- X_{ijk} contains firm controls: age, cash flow, sales, and sales growth;
- λ_k are country fixed effects;
- ω_j are industry fixed effects.

Interpretation of γ_1.

- If access to finance matters, industries that are more dependent on outside funds should have higher investment in countries with better financial-market access.
- If stock market financing is the key margin for R&D, then γ_1 should be positive when $FinDevelop_k$ measures equity-market access.
- If debt is poorly suited for R&D, then γ_1 should be close to zero when $FinDevelop_k$ measures credit-market access.

3.3 Cross-sectional rather than within-firm estimation

- The paper does not estimate the model on annual firm-level panels with firm fixed effects.
- Instead, it uses long-run averages of firm variables over 1990–2007.
- The authors argue that this is especially important for R&D because R&D spending is smooth while stock issuance is lumpy. Year-to-year regressions would mix financing spikes with gradual investment behavior.

Interpretation. The empirical object is a firm's *long-run investment intensity*, not its immediate reaction to one year's financing shock.

3.4 Alternative outcome regressions

The same basic interaction design is estimated with several alternative dependent variables:

$$Y_{ijk} = \delta_0 + \delta_1\big(ExternalDepend_j \times FinDevelop_k\big) + \delta_2 X_{ijk} + \lambda_k + \omega_j + u_{ijk}, \tag{3}$$

where Y_{ijk} is alternatively:

- fixed investment/assets,
- intangible assets/net fixed assets,
- value-added growth,
- productivity growth.

Interpretation. These regressions check the broader real implications of equity-market access and also test whether the stock-market effect is specific to innovative activity rather than to investment in general.

3.5 Instrumental-variables strategy

Because the country-level finance measures may be endogenous, the paper instruments equity-market access using legal variables. Conceptually, the endogenous interaction term is instrumented with interactions between industry external dependence and legal variables:

$$\mathit{ExternalDepend}_j \times \mathit{FinDevelop}_k \quad \text{instrumented by} \quad \mathit{ExternalDepend}_j \times Z_k, \tag{4}$$

where Z_k includes combinations of:

- legal origin,
- contract enforcement,
- anti-self-dealing protection.

Interpretation. Country fixed effects absorb the level of the legal variables, so the identifying IV variation comes from the idea that stronger legal protections matter more for high-external-dependence industries if law works through financial access.

3.6 Firm-level financing regressions

To connect the legal environment directly to how firms fund R&D, the paper also estimates:

$$\mathit{StockIssues}_{ijk} = \alpha_0 + \alpha_1 \mathit{InvestorProtection}_k + \alpha_2 X_{ijk} + \alpha_3 Y_k + \omega_j + \varepsilon_{ijk}, \tag{5}$$

where Y_k is a vector of country controls such as GDP per capita, GDP growth, education, government size, creditor rights, private property protection, inflation, and global growth opportunities.

The paper also regresses the long-run ratio *R&D/cash flow* on investor protections and country equity issues.

Interpretation.

- If law matters through equity financing, stronger shareholder protection should be associated with more stock issuance.
- R&D should also be less tightly linked to internal cash flow when access to external equity is better.

3.7 Standard errors and estimation details

- All regressions are estimated by OLS or 2SLS.
- Standard errors are clustered at the country level.
- Most financial variables are scaled by total assets.
- The paper reports economic significance using Rajan–Zingales-style high-minus-low differentials, comparing the 75th and 25th percentiles of industry dependence and country financial development.

Interpretation. Because the financial-development variables vary at the country level, country clustering is the natural and conservative inference choice.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The baseline identification comes from the Rajan–Zingales logic:

- some industries are technologically more dependent on external finance than others;
- some countries give firms better access to stock market funding than others;
- if equity access matters for innovative investment, the effect should be strongest exactly where external-finance needs are greatest.

With country and industry fixed effects included, the coefficient of interest is identified from the interaction between these two dimensions.

Interpretation. The paper is not asking whether firms in richer countries do more R&D on average. It is asking whether *high-external-dependence industries do relatively more R&D in countries with better stock market access.*

4.2 Why the paper's main comparative-static prediction is powerful

A crucial strength of the design is that it predicts a *different pattern across outcomes and financing types*:

- stock market access should matter for R&D;
- credit access should not matter much for R&D;
- stock market access should matter little for fixed investment;
- credit access may matter somewhat for fixed investment.

Interpretation. This gives the paper an internal falsification test. If all finance variables raised all investment variables in the same way, the paper's specific mechanism would be much less convincing.

4.3 Country and industry fixed effects

- Country fixed effects absorb any time-invariant country-level determinants of investment common to all firms in that country, including broad institutional quality, intellectual property protection, or culture.
- Industry fixed effects absorb global industry differences in investment intensity.
- Firm controls absorb observable differences in internal resources and growth opportunities.

Interpretation. These controls do not solve everything, but they sharply narrow the source of variation to the interaction that the theory predicts.

4.4 Why reverse causality is less problematic here

The key regressors are country-level financial development and legal institutions, not a firm's own financing decision. That substantially weakens the simplest reverse-causality story in which high R&D directly creates better stock market access.

Interpretation. The deeper endogeneity concern is omitted-variable bias at the country level, which is why the IV strategy using legal variables matters.

4.5 External dependence from U.S. data

A standard concern in Rajan–Zingales-style designs is whether the U.S. benchmark measure reflects technology or financing frictions. The paper addresses this by:

- computing the benchmark from U.S. firms that plausibly face relatively frictionless capital markets;
- showing robustness when the dependence measure is recomputed using only young U.S. firms;
- showing robustness when it is recomputed using 1980s U.S. data.

Interpretation. These checks matter because the whole design rests on the claim that the U.S.-based measure captures technological need for external finance rather than country-specific distortions.

4.6 Why legal rules are plausible instruments

The IV strategy relies on a large law-and-finance literature that treats legal origin, enforcement, and minority-investor protection as predetermined drivers of stock market development.

- Common-law countries and countries with stronger contract enforcement tend to have deeper and more accessible stock markets.
- Minority-investor protections should be especially relevant for equity supply because outside shareholders need protection from insider expropriation.
- These legal variables are expected to matter much more for equity access than for debt access.

Interpretation. This is exactly the type of instrument that matches the paper's mechanism. The law should affect R&D because it affects access to stock market funding for externally dependent firms.

4.7 Alternative explanations addressed in the paper

The paper conducts several important robustness checks:

- It sets missing R&D values equal to zero to show the main result is not driven by requiring firms to report R&D.
- It excludes countries with fewer than 50 firms to make sure the results do not come from sparse within-country industry variation.

- It drops the biggest contributor countries—Japan, the United Kingdom, and Canada—to show the results are not dominated by a few markets.
- It uses alternative stock-market access measures, including small-firm equity issuance, listed firms per capita, and access-to-equity indexes.
- It uses alternative debt proxies because private credit-to-GDP may not measure the borrowing capacity of listed firms very well.

Interpretation. The message is that the central finding is not a fragile artifact of one financial proxy or one unusual group of countries.

4.8 What the paper can and cannot distinguish

- The paper is strongest on the claim that stock-market access matters for R&D and that legal investor protection affects R&D through equity finance.
- It is weaker on the exact channel inside the firm: for example, whether stock markets help mainly because of more new issuance, broader investor participation, lower cost of capital, or greater tolerance for risky projects.
- The sample includes listed firms only, so the conclusions apply most directly to firms visible in public markets rather than to all firms in the economy.

Interpretation. This is a carefully identified financing paper, not a complete theory of innovation.

4.9 Relation to the literature

The paper's contribution is to connect several literatures that are often kept separate:

- the law-and-finance literature on investor protection and stock market development;
- the finance-and-growth literature on productivity versus physical investment;
- the R&D financing literature on why debt and innovation fit together poorly;
- the debate on bank-based versus market-based financial systems.

Interpretation. The paper's distinctive move is to say: *the finance-growth channel runs importantly through the financing of risky intangible investment.*

5 Tables and figures

This published paper is essentially **table-driven**; the core empirical evidence is contained in Tables I–IX and the appendix table rather than in main-text figures.

5.1 Table I: variable definitions and measurement

Read:

- Table I defines all firm-level, industry-level, law, and finance variables.

Table I
Description of the Variables

Variable	Description
Firm and industry variables	
R&D	The firm average of R&D scaled by the book value of total assets over the period 1990 to 2007. Firm-year observations are Winsorized at the 1% level prior to computing the average. Data are from Compustat Global and North America.
Fixed investment	The firm average of fixed investment scaled by the book value of total assets over the period 1990 to 2007. Firm-year observations are Winsorized at the 1% level prior to computing the average. Data are from Compustat Global and North America.
Intangible assets	The firm average of the stock of intangible assets scaled by the book value of net fixed assets over the period 1990 to 2007. Firm-year observations are Winsorized at the 1% level prior to computing the average. Data are from Compustat Global and North America.
Value-added growth	The firm average of the annual log change in the sum of operating income plus labor expenses over the period 1990 to 2007. Firm-year observations are Winsorized at the 1% level prior to computing the average. Data are from Compustat Global and North America.
Productivity growth	The firm average of revenue growth minus 0.3*growth in fixed assets minus 0.7*growth in employment over the period 1990 to 2007. Firm-year observations are Winsorized at the 1% level prior to computing the average. Data are from Compustat Global and North America.
Cash flow	The firm average of cash flow scaled by the book value of total assets over the period 1990 to 2007. Cash flow is measured as after-tax income before extraordinary items plus depreciation and amortization plus R&D expense. Firm-year observations are Winsorized at the 1% level prior to computing the average. Data are from Compustat Global and North America.
Stock issues	The firm average of net equity issues scaled by the book value of total assets over the period 1990 to 2007. Net equity issues is equal to the sale of common and preferred stock minus the purchase of common and preferred stock. Firm-year observations are Winsorized at the 1% level prior to computing the average. Data are from Compustat Global and North America.
Sales	The firm average of net sales scaled by the book value of total assets over the period 1990 to 2007. Firm-year observations are Winsorized at the 1% level prior to computing the average. Data are from Compustat Global and North America.
Sales growth	The firm average of the annual log change in net sales over the period 1990 to 2007. Firm-year observations are Winsorized at the 1% level prior to computing the average. Data are from Compustat Global and North America.
Employment	The firm average number of employees over the period 1990 to 2007. Data are from Compustat Global and North America.
ExternalDepend	Industry-level dependence on external finance is fixed investment minus cash from operations divided by fixed investment plus R&D for the median U.S. firm in each two-digit SIC industry. Both reliance on external finance (fixed investment minus cash from operations) and total investment (fixed investment plus R&D) are summed over 1990 to 2007 for each firm before dividing. Data are from Compustat North America.

(*Continued*)

Table I—*Continued*

Variable	Description
Law and finance variables	
Legal origin	A dummy variable equal to one if the country is of common law legal origin. From La Porta et al. (1997).
Anti-self dealing index (ASD)	A measure of legal protection of minority shareholders against expropriation by corporate insiders in 2003, scaled between zero and one with higher values indicating stronger shareholder protection (Djankov et al. (2008)). From La Porta's web page: http://mba.tuck.dartmouth.edu/pages/faculty/rafael.laporta/publications.html.
Enforcement	Enforceability of contracts. The relative degree to which contractual agreements are honored. Scaled from 0 to 10 with higher scores indicating higher enforceability (Djankov et al. (2003)). From La Porta's web page (see above).
Country equity issues	The country average of the firm-level variable *Stock issues*. See above.
Accounting standards	Index of the comprehensiveness of corporate annual reports in 1990 developed by the Center for International Financial Analysis and Research, scaled between 0 and 90, with higher values indicating stronger standards. From Levine (1999).
IPOs/GDP	Value of IPOs over GDP. Averaged over 1996 to 2000. From La Porta's web page (see above).
Country-weighted leverage	Value-weighted average debt-to-assets ratio across all sampled firms in a given country. Data are from Compustat Global and North America.
Credit/GDP	Private credit by deposit money banks to GDP. Averaged over the period 1990 to 2007. Data are from the World Bank's Financial Development and Structure Database (Beck, Demirgüç-Kunt, and Levine (2000) and Beck and Demirgüç-Kunt (2009)).

- The firm-level outcomes are all long-run averages over 1990–2007.
- The key interaction variable, *ExternalDepend*, is measured from U.S. firms at the two-digit SIC level.
- The table also clarifies the distinction between stock-market access measures (country equity issues, accounting standards, IPOs/GDP) and credit-market access measures (country-weighted leverage, credit/GDP).

Economic message. Table I is more than a codebook. It shows that the paper is built to compare *types of finance*, not just the overall level of financial development.

5.2 Table II: sample characteristics

Read:

- Mean R&D/assets is 0.053 in the full sample, very close to mean fixed investment/assets of 0.048.
- Small firms have mean R&D/assets of 0.066 versus 0.024 for large firms.
- Small firms have lower cash flow (0.048 versus 0.104) and much higher equity issues/assets (0.088 versus 0.012).
- The differences in means between small and large firms are all statistically significant.

Economic message. Small firms look exactly like the firms the theory has in mind: more R&D intensive, more reliant on equity, and less able to fund themselves internally.

5.3 Table III: baseline R&D regressions

Read:

- Using *Country equity issues*, the interaction coefficient is 0.413 and highly significant.
- Using *Accounting standards*, the coefficient is 0.253 and significant.
- Using *IPOs/GDP*, the coefficient is 0.454 and significant.
- Using *Country-weighted leverage* or *Credit/GDP*, the coefficients are negative and statistically insignificant.
- The implied differential in R&D intensity is 0.015 under the preferred country-equity-issues measure.

Table II

Sample Characteristics

The firm-level data are from Compustat Global and North America and consist of R&D reporting firms with at least three nonmissing R&D observations during 1990 to 2007. Firms with a primary industry classification in financials (SIC 6000–6999) and utilities (SIC 4900–4999) are excluded. All firm-level variables are averages over the full sample period. Firms are classified as small if their average employment during the sample period is in the bottom 70th percentile of all sampled firms and large otherwise. All variables except *Employment* are scaled by the book value of total assets.

	R&D	Fixed Investment	Cash Flow	Equity Issues	Employment
All firms					
Mean	0.053	0.048	0.064	0.065	5,933
Median	0.018	0.038	0.080	0.005	871
Std dev	0.089	0.038	0.140	0.145	20,443
Small firms					
Mean	0.066	0.044	0.048	0.088	640
Median	0.022	0.033	0.067	0.010	451
Std dev	0.101	0.038	0.159	0.167	586
Large firms					
Mean	0.024	0.056	0.104	0.012	18,145
Median	0.013	0.050	0.097	0.001	6,993
Std dev	0.036	0.035	0.066	0.032	34,176
Tests of difference in means (p-value)	0.000	0.000	0.000	0.000	0.000

Table III

Firm-Level R&D Investment and Access to External Finance

Table III reports OLS regressions with firm-level R&D to assets as the dependent variable. The firm-level data are from Compustat Global and North America and consist of R&D reporting firms with at least three nonmissing R&D observations during 1990 to 2007. All firm-level variables are averages over the full sample period. Firms with a primary industry classification in financials (SIC 6000–6999) and utilities (SIC 4900–4999) are excluded. The key independent variable is the interaction between industry-level dependence on external funds (*ExternalDepend*) and country-level measures of access to external finance. All regressions include firm-level control variables (age, cash flow, sales, and sales growth) as well as country and industry fixed effects. Detailed variable definitions are provided in Table I. The differential in R&D intensity measures the difference in R&D/assets in an industry at the 75th percentile of external dependence relative to an industry at the 25th percentile in a country at the 75th percentile of development versus a county at the 25th percentile. Standard errors calculated with clustering at the country level are in parentheses.

	Financial Development Measured as				
	Country Equity Issues (1)	Accounting Standards (2)	IPOs/ GDP (3)	Country-Weighted Leverage (4)	Credit/ GDP (5)
ExternalDepend × Country equity issues	0.413 (0.097)	–	–	–	–
ExternalDepend × Accounting standards	–	0.253 (0.081)	–	–	–
ExternalDepend × IPOs/GDP	–	–	0.454 (0.181)	–	–
ExternalDepend × Country weighted leverage	–	–	–	– 0.063 (0.201)	–
ExternalDepend × Credit/GDP	–	–	–	–	– 0.011 (0.024)
R^2	0.359	0.352	0.353	0.348	0.348
Observations	5,310	5,244	5,310	5,310	5,310
Differential in R&D	0.015	0.014	0.010	– 0.002	– 0.002

Economic message. The baseline evidence says that external equity access matters for innovative investment, while debt access does not. This is the core result of the paper.

5.4 Table IV: IV regressions using legal rules and institutions

Read:

- Instrumenting country equity issues with *Legal origin* yields a coefficient of 0.359.
- Instrumenting with *Enforcement* yields 0.689.
- Using both legal origin and enforcement yields 0.444.
- Using ASD and enforcement yields 0.445.
- Instrumenting *Accounting standards* and *IPOs/GDP* also produces positive and significant coefficients (0.388 and 0.814, respectively).

Economic message. Once legal rules are used to isolate exogenous variation in stock-market access, the results remain large and significant. This is the paper's main causal evidence that law affects R&D through equity-market development.

5.5 Table V: size and age splits

Read:

- For small firms, the coefficient on *ExternalDepend* × *Country equity issues* is 0.630; for large firms it is only 0.053.
- For young firms, the same coefficient is 0.668; for mature firms it is 0.091.
- Similar small-versus-large and young-versus-mature contrasts appear for accounting standards, IPOs/GDP, and the IV specification.

Table IV

R&D Regressions: Instrument with Legal Rules and Institutions

Table IV reports 2SLS regressions with firm-level R&D to assets as the dependent variable. The key independent variable is the interaction between industry-level dependence on external funds (*ExternalDepend*) and various measures of country-level access to external finance. The instruments for country-level access to external equity finance are: a dummy variable indicating common law legal origin (*Legal origin*), an index measuring the enforcement of contracts (*Enforcement*), and the anti-self-dealing index (*ASD*). All regressions include firm-level control variables (age, cash flow, sales, and sales growth) as well as country and industry fixed effects. The sample is described in Table II. Detailed variable definitions are provided in Table I. The differential in R&D intensity measure is explained in Table III. Standard errors calculated with clustering at the country level are in parentheses.

	Financial Development Measured as					
	Country Equity Issues				Accounting Standards	IPOs/GDP
	(1)	(2)	(3)	(4)	(5)	(6)
Instruments	Legal Origin	Enforcement	Legal Origin and Enforcement	ASD and Enforcement	Legal Origin and Enforcement	Legal Origin and Enforcement
ExternalDepend × Country equity issues	0.359 (0.162)	0.689 (0.275)	0.444 (0.123)	0.445 (0.177)	–	–
ExternalDepend × Accounting standards	–	–	–	–	0.388 (0.161)	–
ExternalDepend × IPOs/GDP	–	–	–	–	–	0.814 (0.386)
R^2	0.359	0.356	0.361	0.361	0.354	0.352
Observations	5,310	5,125	5,125	5,125	5,076	5,125
Differential in R&D	0.013	0.024	0.016	0.016	0.021	0.018

Table V

R&D Regressions: Size and Age Splits

Table V reports OLS and 2SLS regressions for separate samples of small and large firms (Panel A) and young and mature firms (Panel B). Firm-level R&D to assets is the dependent variable. The key independent variable is the interaction between industry-level dependence on external funds (*ExternalDepend*) and country-level measures of access to external finance (*FinDevelop*). All regressions include firm-level control variables (age, cash flow, sales, and sales growth) as well as country and industry fixed effects. In columns (7) and (8), *Legal origin* and *Enforcement* are used as instruments for *Country equity issues*. The sample is described in Table II. Detailed variable definitions are provided in Table I. Firms are classified as small if their average employment during the sample period is in the bottom 70th percentile of all sampled firms and large otherwise. Firms that first appear in Compustat before 1990 are considered "mature," and those appearing after 1990 as "young." The differential in R&D intensity measure is explained in Table III. Standard errors calculated with clustering at the country level are in parentheses. Superscripts ^{a},^{b}, and ^{c} indicate that the coefficient estimate for the small (young) firms is statistically different from the coefficient estimate for the large (mature) firms at the 1%, 5%, and 10% level, respectively.

	Financial Development Measured as							
	Country Equity Issues		Accounting Standards		IPOs/ GDP		Country Equity Issues (IV)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Size Split								
Firms	Small	Large	Small	Large	Small	Large	Small	Large
ExternalDepend × FinDevelop	0.630^a (0.119)	0.053 (0.044)	0.339^b (0.150)	0.021 (0.030)	0.794^b (0.369)	0.073 (0.057)	0.547^a (0.154)	0.036 (0.059)
R^2	0.398	0.433	0.389	0.432	0.391	0.433	0.399	0.437
Observations	3,689	1,621	3,646	1,598	3,689	1,621	3,583	1,542
Differential in R&D	0.022	0.002	0.019	0.001	0.018	0.002	0.019	0.001
Panel B: Age Split								
Firms	Young	Mature	Young	Mature	Young	Mature	Young	Mature
ExternalDepend × FinDevelop	0.668^a (0.124)	0.091 (0.056)	0.287^a (0.092)	0.047 (0.047)	0.698^b (0.295)	0.116 (0.073)	0.629^a (0.162)	0.108 (0.049)
R^2	0.389	0.300	0.379	0.297	0.380	0.299	0.390	0.299
Observations	3,650	1,660	3,599	1,645	3,650	1,660	3,473	1,652
Differential in R&D	0.024	0.003	0.016	0.003	0.016	0.003	0.022	0.004

- The implied differential in R&D intensity for small firms is about 0.022, versus only about 0.002 for large firms.

Economic message. This is the sharpest validation of the mechanism. Equity-market access matters most exactly where external finance should matter most: in small and young firms.

5.6 Table VI: robustness checks

Read:

- Recomputing industry external dependence from young U.S. firms gives a coefficient of 0.367.
- Setting missing R&D values equal to zero still gives a positive and significant coefficient of 0.145.
- Excluding countries with fewer than 50 firms leaves the small-firm coefficient at 0.612 and the large-firm coefficient at 0.069.
- Dropping Japan, the United Kingdom, and Canada leaves the small-firm coefficient at 0.614 and the large-firm coefficient at 0.106.

Economic message. The main R&D result is stable across alternative dependence measures and alternative samples, and the small-firm concentration of the effect survives all of them.

5.7 Table VII: fixed investment regressions

Read:

- The interaction with stock-market access is slightly negative or near zero in all specifications: −0.019, −0.027, and −0.037 for country equity issues, accounting standards, and IPOs/GDP.

Table VI

R&D Regressions: Robustness Checks

Table VI reports OLS regressions with firm-level R&D to assets as the dependent variable. The key independent variable is the interaction between industry-level dependence on external funds (*ExternalDepend*) and *Country equity issues*. All regressions include firm-level control variables (age, cash flow, sales, and sales growth) as well as country and industry fixed effects. The sample is described in Table II. Detailed variable definitions are provided in Table I. Firms are classified as small if their average employment during the sample period is in the bottom 70^{th} percentile of all sampled firms and large otherwise. In column (1), *ExternalDepend* measures industry external financial dependence over 1990–2007 using only U.S. firms that are within 10 years since they first appear in Compustat. In column (2), we set R&D to zero for any firm that reports information on fixed investment but not R&D. In columns (3) and (4), we exclude all countries with less than 50 firms in the original sample (roughly the median firm count). In columns (5) and (6), we drop the three largest contributor countries in terms of firms: Japan, the United Kingdom, and Canada. The differential in R&D intensity measure is explained in Table III. Standard errors are calculated with clustering at the country level and presented in parentheses. Superscripts [a], [b], and [c] indicate that the coefficient estimate for the small firms is statistically different from the coefficient estimate for the large firms at the 1%, 5%, and 10% level, respectively.

	Robustness Check					
	Dependence Young U.S. Firms	Set Missing R&D = 0	Excl. Countries with Less than 50 Firms		Excl. Japan, United Kingdom, & Canada	
	(1)	(2)	(3)	(4)	(5)	(6)
Firms:	All	All	Small	Large	Small	Large
ExternalDepend × Country equity issues	0.367	0.145	0.612[a]	0.069	0.614[b]	0.106
	(0.086)	(0.045)	(0.122)	(0.058)	(0.265)	(0.081)
R^2	0.358	0.281	0.394	0.431	0.337	0.450
Observations	5,307	13,557	3,444	1,436	1,315	734
Differential in R&D	0.013	0.005	0.022	0.002	0.022	0.004

Table VII

Firm-Level Fixed Investment and Access to External Finance

Table VII reports OLS regressions with firm-level fixed investment to assets as the dependent variable. The key independent variable is the interaction between industry-level dependence on external funds (*ExternalDepend*) and country-level measures of access to external finance. All regressions include firm-level control variables (age, cash flow, sales, and sales growth) as well as country and industry fixed effects. The sample is described in Table II. Detailed variable definitions are provided in Table I. The differential in fixed investment intensity measures the difference in fixed investment/assets in an industry at the 75^{th} percentile of external dependence relative to an industry at the 25^{th} percentile in a country at the 75^{th} percentile of development versus a country at the 25^{th} percentile. Standard errors are calculated with clustering at the country level and presented in parentheses.

	Financial Development Measured as				
	Country Equity Issues (1)	Accounting Standards (2)	IPOs/ GDP (3)	Country-Weighted Leverage (4)	Credit/ GDP (5)
ExternalDepend × Country equity issues	– 0.019	–	–	–	–
	(0.021)				
ExternalDepend × Accounting standards	–	– 0.027	–	–	–
		(0.013)			
ExternalDepend × IPOs/GDP	–	–	– 0.037	–	–
			(0.028)		
ExternalDepend × Country weighted leverage	–	–	–	0.056	–
				(0.019)	
ExternalDepend × Credit/GDP	–	–	–	–	0.003
					(0.003)
R^2	0.272	0.274	0.272	0.272	0.272
Observations	5,292	5,226	5,292	5,292	5,292
Differential in fixed inv.	– 0.001	– 0.001	– 0.001	0.002	0.001

- The interaction with country-weighted leverage is positive and significant at 0.056.
- The implied fixed-investment differential under leverage is only about 0.002, or about 4% of average fixed investment intensity.

Economic message. This table is the key contrast result. Stock markets matter for R&D, not for fixed capital. Debt matters modestly for fixed capital, not for R&D. That pattern is exactly what the theory predicts.

5.8 Table VIII: intangible assets and firm growth

Read:

- For small firms, the coefficient linking stock-market access to intangible-assets intensity is 7.897; for large firms it is −0.850.
- For small firms, the coefficient on value-added growth is 1.441; for large firms it is −0.159.
- For small firms, the coefficient on productivity growth is 0.252; for large firms it is −0.030.
- The implied differentials are economically meaningful: about 0.279 for intangible-assets intensity, 0.051 for value-added growth, and 0.009 for productivity growth in the small-firm samples.

Economic message. Equity-market access does not just raise R&D spending. It is associated with more intangible capital and faster growth where innovative activity should be most important.

5.9 Table IX: legal rules and financing R&D

Read:

- Panel A shows that stronger investor protection predicts more stock issuance. The ASD coefficient is 0.129 for small firms but only 0.012 for large firms. Using legal origin instead of ASD gives

Table VIII

Intangible Assets, Firm Growth, and Access to External Finance

Table VIII reports OLS regressions where the dependent variable is firm-level intangible assets/net fixed assets in columns (1) and (2), firm-level value-added growth in columns (3) and (4), and firm-level productivity growth in columns (5) and (6). The key independent variable is the interaction between industry-level dependence on external funds (*ExternalDepend*) and *Country equity issues*. The sample is described in Table II. Detailed variable definitions are provided in Table I. The firm size split is described in Table V. The differential in dependent variable measures the difference in the dependent variable in an industry at the 75[th] percentile of external dependence relative to an industry at the 25[th] percentile in a country at the 75[th] percentile of development versus a country at the 25[th] percentile. The sample average is the average value of the dependent variable in a particular regression sample. The regressions in columns (1) and (2) include firm-level control variables (age, cash flow, sales, and sales growth) and the regressions in columns (3)–(6) include firm-level control variables age and initial size (number of employees the year the firm first enters the sample). All regressions include country and industry fixed effects. Standard errors are calculated with clustering at the country level and presented in parentheses. Superscripts [a], [b], and [c] indicate that the coefficient estimate for the small firms is statistically different from the coefficient estimate for the large firms at the 1%, 5%, and 10% level, respectively.

	Dependent Variable					
	Intangible Assets		Value-Added Growth		Productivity Growth	
Firms	(1) Small	(2) Large	(3) Small	(4) Large	(5) Small	(6) Large
ExternalDepend × Country equity issues	7.897[b]	− 0.850	1.441[a]	− 0.159	0.252[c]	− 0.030
	(3.943)	(1.244)	(0.284)	(0.250)	(0.124)	(0.191)
R^2	0.224	0.241	0.090	0.218	0.068	0.161
Observations	3,665	1,616	3,189	1,615	3,451	1,591
Differential in dependent variable	0.279	− 0.030	0.051	− 0.006	0.009	− 0.001
Sample average	1.368	0.410	0.130	0.099	0.023	0.018

Table IX

Legal Rules and Financing R&D

Panel A in Table IX reports OLS regressions with firm-level stock issues to assets as the dependent variable. In Panel B, firm-level R&D to internal cash flow is the dependent variable. The sample is described in Table II. Detailed variable definitions are provided in Table I. The firm size split is described in Table V. All regressions in Table IX include the following country control variables: log (GDP/Pop), GDP growth, secondary school enrollment, creditor rights, private property rights protection, global growth opportunities, government expenditure/GDP, and inflation rate. Regressions in Panel A also include firm-level control variables (age, cash flow, sales, and sales growth) as well as industry fixed effects; regressions in Panel B include the same controls less firm cash flow. Standard errors are calculated with clustering at the country level and presented in parentheses. Superscripts [a], [b], and [c] indicate that the coefficient estimate for the small firms is statistically different from the coefficient estimate for the large firms at the 1%, 5%, and 10% level, respectively.

Panel A: Law and External Equity Issues

	Dependent Variable: Stock Issues/Assets					
Firms	All (1)	Small (2)	Large (3)	All (4)	Small (5)	Large (6)
ASD	0.119	0.129[a]	0.012	–	–	–
	(0.029)	(0.031)	(0.004)			
Legal origin	–	–	–	0.071	0.079[a]	0.005
				(0.009)	(0.008)	(0.002)
R^2	0.365	0.392	0.203	0.374	0.401	0.202
Observations	5,185	3,616	1,569	5,185	3,616	1,569

Panel B: Law and R&D Relative to Cash Flow

	Dependent variable: R&D/cash flow					
Firms	All (1)	All (2)	All (3)	All (4)	Small (5)	Large (6)
ASD	0.310	–	–	0.019	0.340[a]	− 0.147
	(0.111)			(0.099)	(0.146)	(0.045)
Legal origin	–	0.204	–	–	–	–
		(0.054)				
Country equity issues	–	–	2.449	2.420	–	–
			(0.356)	(0.501)		
R^2	0.211	0.213	0.215	0.215	0.254	0.261
Observations	4,887	4,887	4,887	4,887	3,323	1,564

0.079 for small firms and 0.005 for large firms.

- Panel B shows that R&D relative to cash flow is higher in countries with stronger investor protection and higher country equity issuance.
- Once ASD and country equity issues are included together, the country-equity-issues coefficient remains strongly positive while ASD itself becomes very small.
- The positive relation between investor protection and R&D/cash flow is driven by small firms.

Economic message. This table ties the whole story together. Law affects stock issuance, and stock issuance relaxes the dependence of R&D on internal cash flow. That is exactly the channel the paper wants to establish.

5.10 Appendix Table A.I: country characteristics

Appendix: Country Characteristics

Table A.I

This table reports firm counts and investor protection and financial development measures for all sampled countries. All variables and data sources are listed in Table I.

Country	Firm Count	Legal Origin	Anti–self Dealing Index (ASD)	Enforcement	Country Equity Issues	Accounting Standards	IPOs/ GDP	Country-Weighted Leverage	Credit/ GDP
Australia	188	Common	0.79	9.36	0.197	75	0.087	0.280	0.793
Austria	29	Civil	0.21	9.80	0.037	54	0.012	0.287	0.979
Belgium	39	Civil	0.54	9.74	0.088	61	0.024	0.276	0.693
Brazil	24	Civil	0.29	6.31	0.007	54	0.001	0.199	0.327
Canada	484	Common	0.65	9.48	0.175	74	0.086	0.272	0.933
Chile	12	Civil	0.63	6.91	0.013	52	0.005	0.235	0.534
Denmark	41	Civil	0.47	9.66	0.117	62	0.012	0.240	0.864
Finland	83	Civil	0.46	9.58	0.029	77	0.038	0.282	0.673
France	169	Civil	0.38	9.09	0.045	69	0.023	0.257	0.885
Germany	250	Civil	0.28	9.50	0.075	62	0.028	0.263	1.072
Greece	51	Civil	0.23	6.40	0.014	55	0.088	0.274	0.454
Hong Kong	53	Common	0.96	n/a	0.105	69	0.091	0.190	1.457
India	243	Common	0.55	5.14	0.026	57	0.006	0.217	0.275
Indonesia	17	Civil	0.68	n/a	0.031	n/a	0.017	0.304	0.344
Ireland	36	Common	0.79	8.38	0.112	n/a	0.061	0.287	0.896
Israel	130	Common	0.71	6.18	0.132	64	0.004	0.222	0.703
Italy	32	Civil	0.39	8.75	0.045	62	0.059	0.338	0.680
Japan	2,167	Civil	0.48	9.34	0.013	65	0.024	0.284	1.540
Korea	19	Civil	0.46	6.97	0.063	62	0.053	0.317	0.680
Malaysia	83	Common	0.95	7.11	0.050	76	0.062	0.268	1.112
Netherlands	70	Civil	0.21	9.68	0.096	64	0.026	0.209	1.179
New Zealand	13	Common	0.95	9.65	0.054	70	0.001	0.335	1.019
Norway	43	Civil	0.44	9.86	0.096	74	0.022	0.232	0.641
Pakistan	13	Common	0.41	3.95	0.000	n/a	0.004	0.185	0.233

(*Continued*)

Table A.I—*Continued*

Country	Firm Count	Legal Origin	Anti–self Dealing Index (ASD)	Enforcement	Country Equity Issues	Accounting Standards	IPOs/ GDP	Country-Weighted Leverage	Credit/ GDP
Philippines	17	Civil	0.24	3.77	0.015	65	0.022	0.166	0.306
Singapore	73	Common	1.00	n/a	0.051	78	0.059	0.278	0.935
South Africa	45	Common	0.81	5.85	0.019	70	0.007	0.168	0.618
Spain	11	Civil	0.37	8.10	0.008	64	0.024	0.362	0.956
Sweden	133	Civil	0.34	9.79	0.108	83	0.063	0.231	0.640
Switzerland	107	Civil	0.27	9.99	0.048	68	0.071	0.237	1.592
Turkey	43	Civil	0.43	n/a	0.009	51	0.015	0.251	0.155
United Kingdom	643	Common	0.93	9.10	0.133	78	0.113	0.206	1.254
Mean			**0.54**	**8.12**	**0.063**	**66**	**0.038**	**0.255**	**0.795**
Median			**0.47**	**9.09**	**0.049**	**65**	**0.024**	**0.260**	**0.748**

Read:

- The appendix reports country counts and the full set of legal and financial development measures.
- There is large variation in anti-self-dealing protection, enforcement, equity issuance, accounting standards, leverage, and credit depth.
- The Netherlands and South Africa are natural 75th/25th percentile comparison countries for country equity issues in the paper's differential calculations.
- Japan contributes by far the largest number of firms, which explains why the paper later checks that results survive dropping it.

Economic message. The appendix shows that the identifying country-level variation is economically substantial, not just statistically detectable.

6 Short summary

- The paper asks whether legal rules and financial-market structure affect firm-level innovative investment.
- Its main idea is that stock market funding is especially important for R&D because R&D is intangible, risky, and hard to collateralize.
- Using a Rajan–Zingales-style interaction design on about 5,300 firms from 32 countries, the paper finds that better stock market access raises long-run R&D investment, especially in small and young firms.
- Credit market development has no effect on R&D, but a modest effect on fixed investment.
- Stock market access also predicts more intangible assets and faster value-added and productivity growth for small firms.
- Instrumenting stock-market access with legal rules such as legal origin, contract enforcement, and anti-self-dealing protection gives similar results.
- Stronger investor protections are associated with more stock issuance and with R&D that is less tied to internal cash flow, supporting the external-equity-financing mechanism.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that legal investor protections and stock market development have a substantial causal effect on firm-level innovative investment. The effect is strongest for small and young firms, is specific to equity-market access rather than debt-market depth, and is concentrated in R&D and other intangible-growth margins rather than in fixed capital spending.

7.2 Economic intuition

When firms invest in assets that can be pledged as collateral, weak equity markets can often be offset by debt. But R&D is different: it is intangible, risky, and disproportionately driven by upside outcomes that debt contracts do not capture well. Because of this, legal rules and financial developments that widen access to outside equity should matter greatly for innovation, while having much smaller effects on ordinary fixed investment.

7.3 Takeaway

The paper's message is not simply that *finance matters*. It is that the *type of finance* matters, and that legal institutions matter most when firms need external equity to fund risky intangible investment. That insight provides a concrete firm-level channel linking law, stock markets, innovation, and long-run productivity growth.